AI Assignment #2

CISC3024 Pattern Recognition
University of Macau

Monte Carlo Dropout as a Bayesian Approximation
for Fashion-MNIST Image Classification

October 2026

Algorithm: Gal & Ghahramani, ICML 2016 (arXiv:1506.02142)

Source code: https://github.com/LeiManLam/CISC3024-AI-Assignment2

1. How the AI Finds the Algorithm

The assignment requires a recent classification/recognition method based on probability or Bayes theory for computer vision / pattern recognition, with the AI performing algorithm search, programming, and report writing.

Search process. Queries used included: “Bayesian neural network computer vision classification 2020–2023 MC Dropout variational inference”, “Monte Carlo Dropout Bayesian approximation Gal Ghahramani image classification”, and “Mobiny Monte Carlo DropConnect Bayesian deep networks”. Sources: arXiv, NeurIPS proceedings, and Scientific Reports.

Candidates considered:

- MC Dropout (Gal & Ghahramani, ICML 2016, arXiv:1506.02142) — dropout as approximate variational inference in deep Gaussian processes; still a standard Bayesian DL baseline in 2020+ vision and medical-imaging benchmarks.
- MC-DropConnect (Mobiny et al., Scientific Reports, 2021; arXiv:1906.04569) — Bernoulli variational approximation on weights; validated on classification and segmentation.
- Variational Structured Dropout (VSD) (NeurIPS 2021) — structured variational noise; stronger theory but heavier to implement on CPU.
- Function-space VI / MC-DropBlock (NeurIPS 2022 / 2021) — excellent uncertainty work but more complex architectures and compute.

Selection. We chose Monte Carlo Dropout because (1) it has a clear Bayes / variational-inference foundation (approximate posterior over weights via Bernoulli dropout masks), (2) it applies directly to CNN image classification, (3) it remains widely used after 2020 (e.g. NeurIPS 2021 Bayesian DL benchmarks), and (4) it runs on CPU with Fashion-MNIST. MC-DropConnect (2021) was a strong recent alternative; MC Dropout was preferred for simpler reproducibility while keeping the same Bayesian interpretation.

2. Algorithm Description

Reference: Yarin Gal and Zoubin Ghahramani. “Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning.” ICML, 2016, pp. 1050–1059. arXiv:1506.02142.

Core idea. Training a neural network with dropout (and L2 weight decay) can be interpreted as approximate variational inference for a deep Gaussian process. The variational distribution q(ω) places Bernoulli masks on network units. Minimising the dropout objective approximates minimising KL[q(ω) || p(ω | data)].

Predictive inference (MC Dropout). At test time, dropout stays active. We draw T stochastic forward passes with independent masks ω _t ~ q(ω) and average the softmax probabilities:

$$q(y^*|x^*) \approx (1/T) \Sigma_t \text{Softmax}(f_{\omega\ _t}(x^*))$$

From the averaged probabilities we obtain the predicted class, negative log-likelihood (NLL), and uncertainty summaries such as predictive entropy. This is a probability / Bayes-theory approach for computer vision classification that also indicates when the model is unsure.

Why it fits this course. Pattern recognition historically uses Bayes classifiers. MC Dropout brings that Bayesian predictive view to modern CNNs: visual features are learned as usual, but predictions come from an approximate posterior rather than a single point-estimate weight vector.

3. How the AI Implements the Algorithm

All code was written by the AI agent (Python + PyTorch). The student did not write source code.

- model.py — BayesianCNN: two conv blocks (32→64), max-pooling, FC 128→10, Dropout2d/Dropout (p=0.25). Method mc_predict(x, T) keeps dropout stochastic and averages T softmax outputs.
- train_eval.py: Fashion-MNIST, Adam (lr=1e-3, weight decay=1e-4), 8 epochs, batch 128; evaluates deterministic (dropout off) and MC Dropout (T=30).
- metrics.py: accuracy, mean NLL, Expected Calibration Error (ECE, 15 bins), confusion matrix.
- Figures: training curves, confusion matrix, reliability diagram, entropy histograms (correct vs incorrect).

Public source: https://github.com/LeiManLam/CISC3024-AI-Assignment2.

4. Experiment Settings and Results

Dataset. Fashion-MNIST: 60,000 train + 10,000 test grayscale 28×28 images, 10 clothing classes.

Setting	Value
Algorithm	MC Dropout (Gal & Ghahramani, 2016)
Model	CNN (~422K parameters)
Dropout p	0.25
MC samples T	30
Epochs / batch	8 / 128
Optimizer	Adam, lr=0.001, wd=0.0001
Device / seed	cpu / 42

Metric	Value
Deterministic test accuracy	91.11%
MC Dropout test accuracy (T=30)	91.07%
MC Dropout mean NLL	0.2450
Expected Calibration Error (ECE)	0.0199
Mean predictive entropy (all)	0.2968
Mean entropy (correct)	0.2385
Mean entropy (incorrect)	0.8910

Discussion. MC Dropout accuracy (~91.07%) matches the deterministic CNN (~91.11%). ECE ≈ 0.020 indicates reasonably calibrated probabilities. Incorrect predictions have much higher predictive entropy (0.891) than correct ones (0.239), so the Bayesian approximation flags uncertain cases.

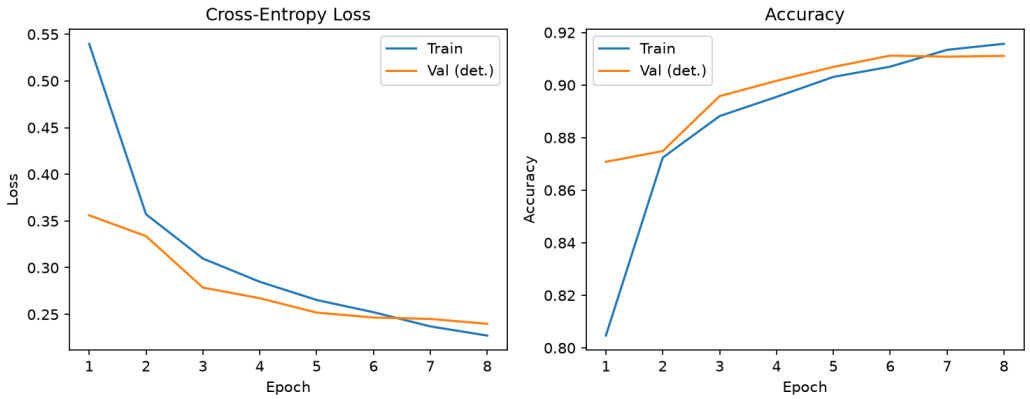


Figure 1. Training / validation loss and accuracy.

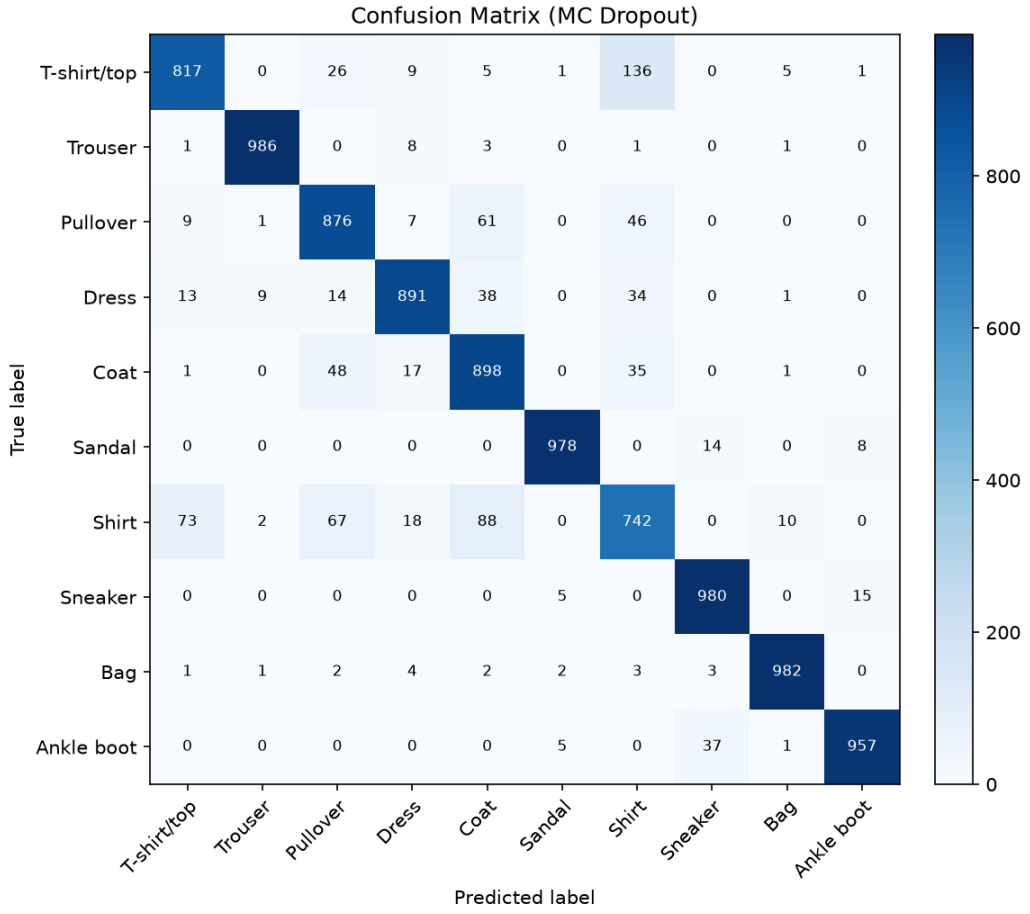


Figure 2. Confusion matrix (MC Dropout, T=30).

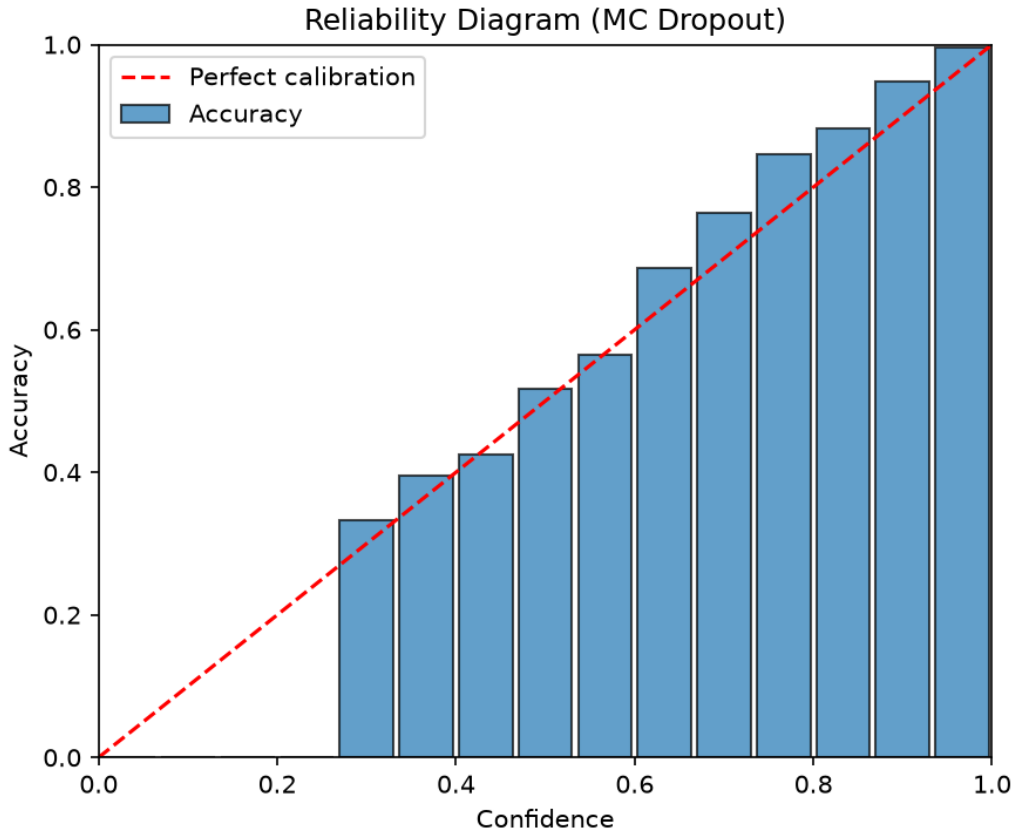


Figure 3. Reliability diagram (calibration).

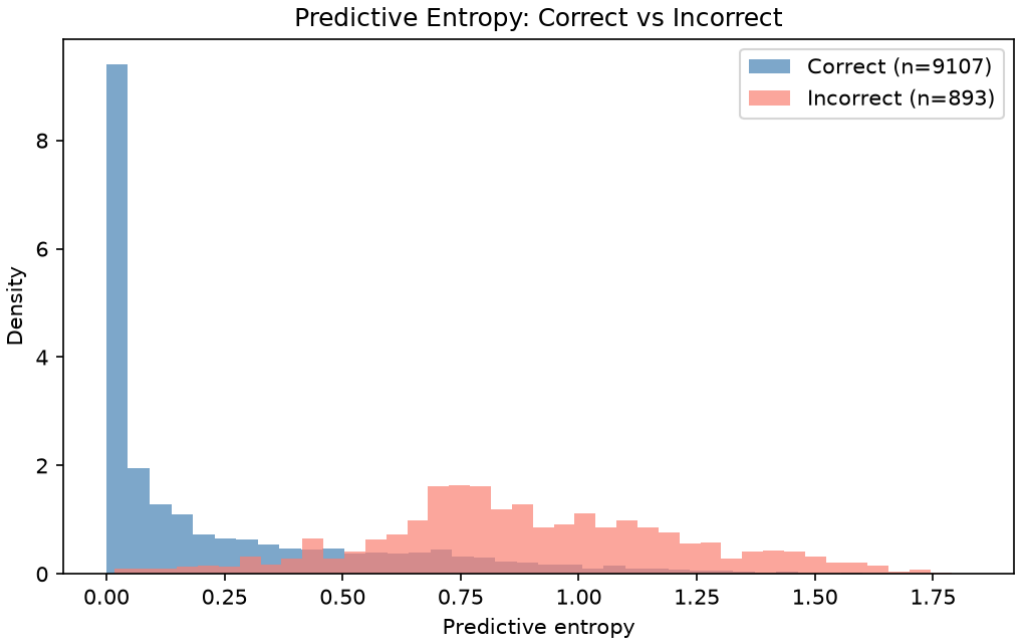


Figure 4. Predictive entropy: correct vs incorrect predictions.

5. What the Student Has Learned from This AI Assignment

（本節以學生第一人稱、繁體中文撰寫。）

透過這次 AI 作業，我學到如何把課堂上的貝氏／機率概念連接到現代深度學習。以前接觸貝氏分類器時，多半是較簡單的生成模型；這次用 Monte Carlo Dropout，才理解「對權重做近似後驗推論」也可以套在

CNN 上，讓影像分類不只給出類別，還能估計模型有多不確定。

我也體會到「查論文」與「能實作」之間的差別。AI 幫我搜尋 2020 年後仍常用的貝氏深度學習方法，比較 MC Dropout、MC-DropConnect、結構化變分 Dropout 等候選後，選了理論清楚、資源需求適中的方案，並在 Fashion-MNIST 上跑出真實的準確率、NLL、ECE 與混淆矩陣，而不是虛構數字。

最重要的是，作業規定學生不可自行寫程式，整段實作與報告由 AI 完成。我學到如何清楚下達需求、檢查實驗設定是否合理、解讀不確定度指標（例如錯誤樣本的預測熵明顯較高），以及把原始碼整理成可公開的專案。這讓我更明白在模式識別中，機率思維不只是公式，而是可以落地成可重現實驗的工程流程。

6. Webpage Link to the Source Codes

All source code, experiment scripts, figures, and this report are published at:

https://github.com/LeiManLam/CISC3024-AI-Assignment2

The repository includes src/model.py, src/train_eval.py, src/metrics.py, requirements.txt, README.md, results/, figures/, and report/.

References

[1] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning,” ICML, 2016. https://arxiv.org/abs/1506.02142

[2] A. Mobiny et al., “DropConnect is effective in modeling uncertainty of Bayesian deep networks,” Scientific Reports, 2021. https://arxiv.org/abs/1906.04569

[3] H. Xiao, K. Rasul, and R. Vollgraf, “Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms,” arXiv:1708.07747, 2017.